Dual-Channel Regularized-Determinant Coherence
Without Cross-Scale Contraction

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Abstract

The full Frobenius complement bound fails, but the shell-complete selected clouds still define finite second-regularized determinant factors. This paper compares those factors without branch matching.

On the closed unit disk all selected resonances satisfy $|z\lambda| < 1$, so the principal logarithm

$$L_\Lambda(z) = \sum_{\lambda\in\Lambda}[\log(1 - z\lambda) + z\lambda]$$

is single valued and permutation invariant. It is evaluated on a 256-point grid for sixteen left/right channel pairs and thirty adjacent-scale pairs.

All sixteen channel comparisons pass the predeclared sup-log gate 0.02; the maximum difference is 0.017855. Adjacent-scale differences are much larger, ranging from 0.042131 to 0.183660. Neither channel is strictly contracting over its last four transitions, and the final adjacent errors are approximately 0.0712 and 0.0792.

Thus the two physical channels remain coherent at each frozen scale, but the selected factors do not form a visibly Cauchy small-noise family. This is a mixed finite result. It supports a shared dual-channel determinant renormalization while ruling out simple convergence of the selected factor as the current mechanism.

1 Branch-free determinant comparison

Individual nonnormal eigenvalue branches can collide and exchange labels. A symmetric spectral function avoids that ambiguity. For a finite multiset Λ, define

$$L_\Lambda(z) = \sum_{\lambda\in\Lambda}[\log(1 - z\lambda) + z\lambda]. \tag{1}$$

Whenever $|z| \max_{\lambda\in\Lambda} |\lambda| < 1$, every logarithm is the branch analytic from zero and

$$e^{L_\Lambda(z)} = \prod_{\lambda\in\Lambda}(1 - z\lambda)e^{z\lambda}.$$

Proposition 1.1 (Permutation and conjugation invariance)**.** *The function L_Λ is invariant under every permutation of the multiset. If Λ is conjugate closed, then*

$$\overline{L_\Lambda(\overline{z})} = L_\Lambda(z).$$

Proof. Both statements follow by permuting or conjugating the summands in (1). □

No cross-level root assignment is required.

2 Trace-power interpretation

Inside the same disk,

$$L_\Lambda(z) = -\sum_{m=2}^{\infty} \frac{z^m}{m}\tau_m(\Lambda), \qquad \tau_m(\Lambda) = \sum_{\lambda\in\Lambda} \lambda^m. \tag{2}$$

Hence comparison of two selected factors measures all finite-cloud power-trace differences simultaneously. The missing $m = 1$ term suppresses barycentric drift but does not erase genuine higher-order spectral motion.

At fixed positive noise, the corresponding infinite trace series is the Hilbert–Schmidt determinant of RH-7 [2]. Here (2) is used only for selected finite clouds.

3 Common grid and gates

The grid consists of 64 angles on each radius

$$0.25, \quad 0.5, \quad 0.75, \quad 1,$$

for 256 points in total. The largest selected resonance modulus in the whole atlas is 0.86860, so the closed unit disk remains strictly inside the logarithmic convergence disk.

For two clouds define

$$d_{\mathcal{G}}(\Lambda, \Gamma) = \max_{z\in\mathcal{G}} |L_\Lambda(z) - L_\Gamma(z)|. \tag{3}$$

The dual-channel gate is frozen at

$$d_{\mathcal{G}}(\Lambda_{\sigma,L}, \Lambda_{\sigma,R}) < 0.02.$$

No gate is imposed after seeing the adjacent-scale values. Their diagnostic is whether the last four transition errors are strictly decreasing.

4 Dual-channel result

All sixteen channel pairs pass. Representative values are:

σ	channel sup-log difference
0.04000	0.003309
0.02500	0.009120
0.01600	0.014354
0.00800	0.017855
0.00500	0.003103
0.00250	0.008517
0.00125	0.014933

The sequence is not monotone, but every value stays below the declared finite gate. This extends the dual-channel coherence of the quartet divisor to rank-growing selected regularized factors.

The ranks differ by one at several scales, so the agreement is not a trivial comparison of identical cardinalities. Second regularization and the small inner-shell moduli reduce the effect of those extra roots.

5 Adjacent-scale result

For each channel, compare consecutive noise levels using the same metric (3). Across 30 cases,

$$0.042131 \le d_{\mathcal{G}} \le 0.183660.$$

These errors are all larger than the worst same-scale channel difference.

The final four left-channel transition errors are approximately

$$0.12033, \quad 0.06780, \quad 0.05237, \quad 0.07121,$$

and the right-channel values are

$$0.11715, \quad 0.08456, \quad 0.04213, \quad 0.07916.$$

Both sequences increase on the final transition. Therefore neither passes strict tail contraction.

diagnostic	result
dual-channel cases	16
dual-channel gate passes	16
maximum channel error	0.017855
adjacent-scale cases	30
adjacent error range	0.042131–0.183660
last-four contraction, left	no
last-four contraction, right	no

6 Interpretation

The mixed verdict rules out two simplistic conclusions.

First, channel coherence does not imply scale convergence. The Haar-compressed and fine endpoints approximate one physical construction at a fixed noise, whereas changing σ changes the operator and increases selected rank.

Second, noncontraction of the selected factor does not disprove convergence of a relative determinant. A moving near-unit cloud may need to be divided out, and the unresolved complement may contribute compensating terms. RH-80 gives the abstract factorization target [3]; RH-229 shows why a raw Frobenius bound cannot supply its uniform complement estimate [1].

7 Claim boundary

The comparison is finite and grid based. Although (1) is analytic on the whole unit disk, a grid maximum is only a sampled diagnostic, not a validated supremum. The products include selected clouds only and omit the unresolved determinant tail.

The justified conclusions are finite dual-channel coherence and absence of the tested cross-scale contraction. No locally uniform determinant, dynamical limit, self-adjoint generator, counting law, or arithmetic trace formula is proved. Gate A remains open.

References

[1] Liang Wang. The nonnormal frobenius tail-budget barrier. 2026. RH-229 preprint.

[2] Liang Wang. Irreversible gaussian cycle spectrum. 2026. RH-7 preprint.

[3] Liang Wang. Moving-cloud renormalization and the relative determinant gate. 2026. RH-80 preprint.